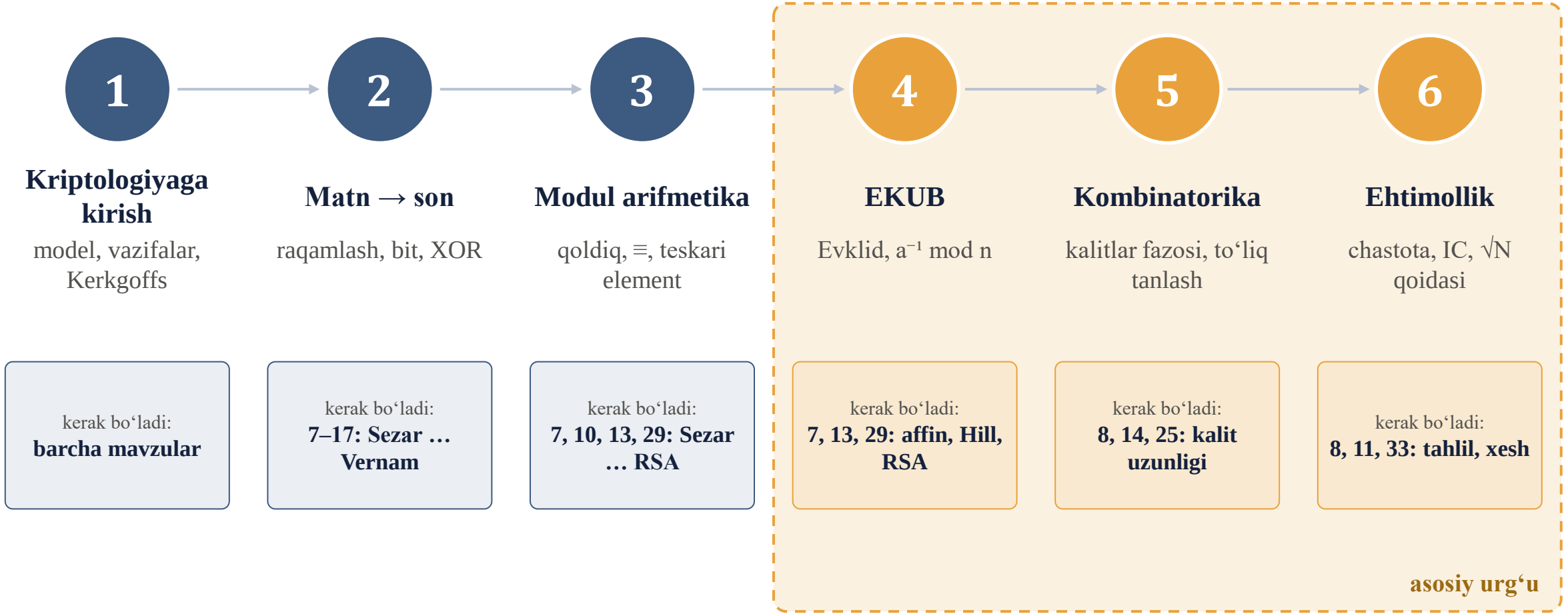


Kriptologiya asoslari

I blok · Poydevor: shifrlarni yaratish va ochishning matematik vositalari

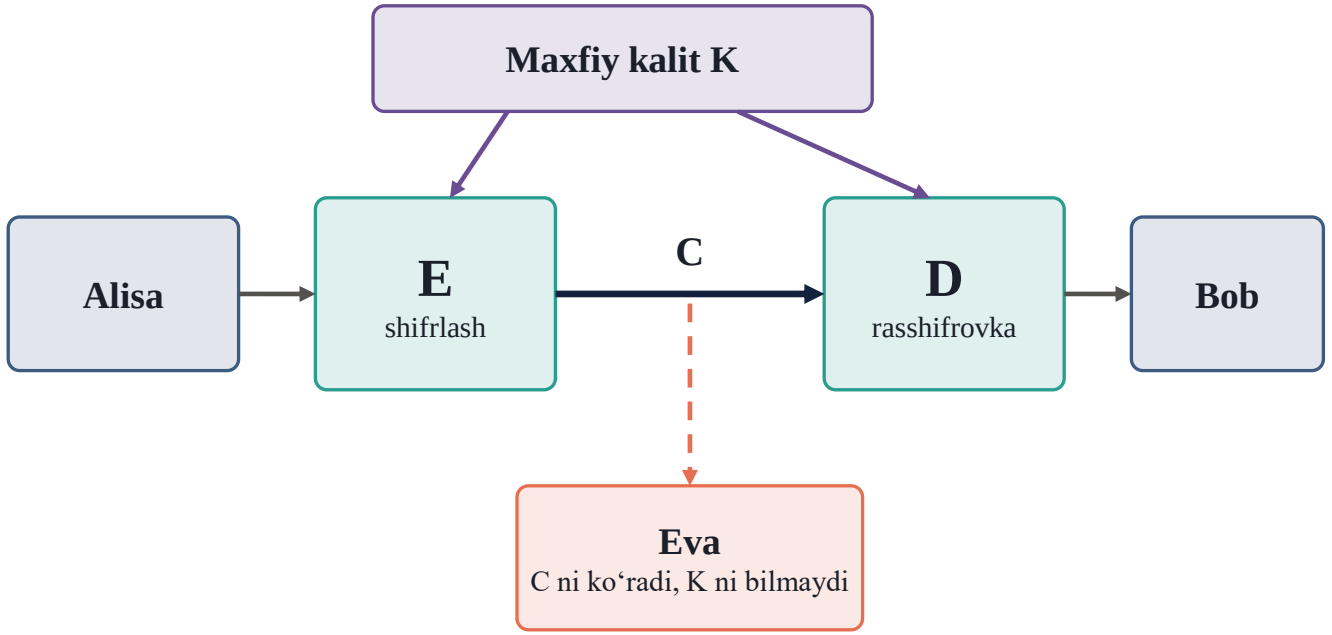
Modul arifmetika · EKUB va teskari element · Kombinatorika · Ehtimollik va statistika

Olti mavzu — oltita vosita



Raqamlar — keyingi bloklardagi mavzu raqamlari: har bir vosita aynan qaysi shifr yoki hujumda ishlatilishi.

Model, prinsip va hujumchi



$$C = E_K(P), \quad P = D_K(C)$$

Misol — Sezar, K = 3: **HELLO** → **KHOOR**

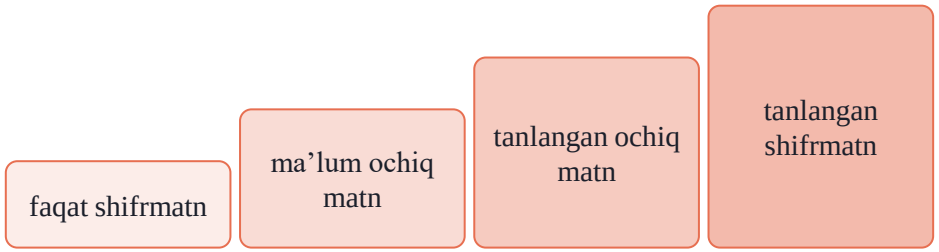
Kerkgoffs prinsipi (1883)

Algoritm raqibga ma’lum deb hisoblanadi. Xavfsizlik faqat kalitning sirligiga tayanadi.

Simmetrik ↔ asimmetrik

Bitta umumiy kalit (AES) ↔ ochiq/yopiq kalitlar jufti (RSA). Amalda: asimmetrik usul bilan kalit kelishiladi, xabar simmetrik shifr bilan shifrlanadi.

Hujumchi kuchi zinasi



Kriptografiyaning asosiy maqsadlari

1. Maxfiylik

конфиденциальность / confidentiality

Savol:

Begona xabarni o‘qiy olmaydimi?

Misol:

Tahlil natijasini faqat bemor ko‘radi

Vosita:

Shifrlash (7–32)

2. Butunlik

целостность / integrity

Savol:

Xabar yo‘lda o‘zgarmadimi?

Misol:

“100 so‘m” “100 000 so‘m”ga aylanmasin

Vosita:

Xesh, MAC (33–34)

3. Autentifikatsiya

аутентификация / authentication

Savol:

Xabarni haqiqatan kim yubordi?

Misol:

“Bankdan” kelgan SMS firibgarnikimi?

Vosita:

MAC, imzo (34–35)

4. Rad eta olmaslik

неотказуемость / non-repudiation

Savol:

Muallif “men yubormaganman” deya oladimi?

Misol:

Elektron imzoli shartnoma

Vosita:

Raqamli imzo (35)

Yordamchi vazifa: kalitlarni boshqarish (управление ключами / key management)

Izoh: bu maqsad emas, to‘rt maqsadning ishlashi uchun shart: ular kalit tasodifiy yaratilgan va xavfsiz yetkazilgan bo‘lsagina ta’minlanadi. Vositalar: tasodifiy sonlar generatori (17), Diffi–Xellman (30), PKI (35).

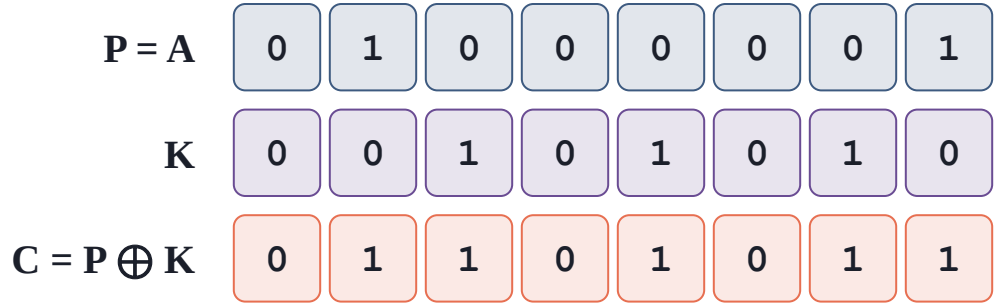
Ikki yondashuv — bitta g‘oya: modul bo‘yicha qo‘shish

Klassik: harf → 0…25, qo‘shish mod 26



Chegara: X = 23 → 23 + 3 = 26 ≡ 0 → A (“aylanib qaytish” — 3-mavzu)

Zamonaviy: bayt → bitlar, XOR = qo‘shish mod 2



C ⊕ K = P
shifrlash va rasshifrovka — bitta amal

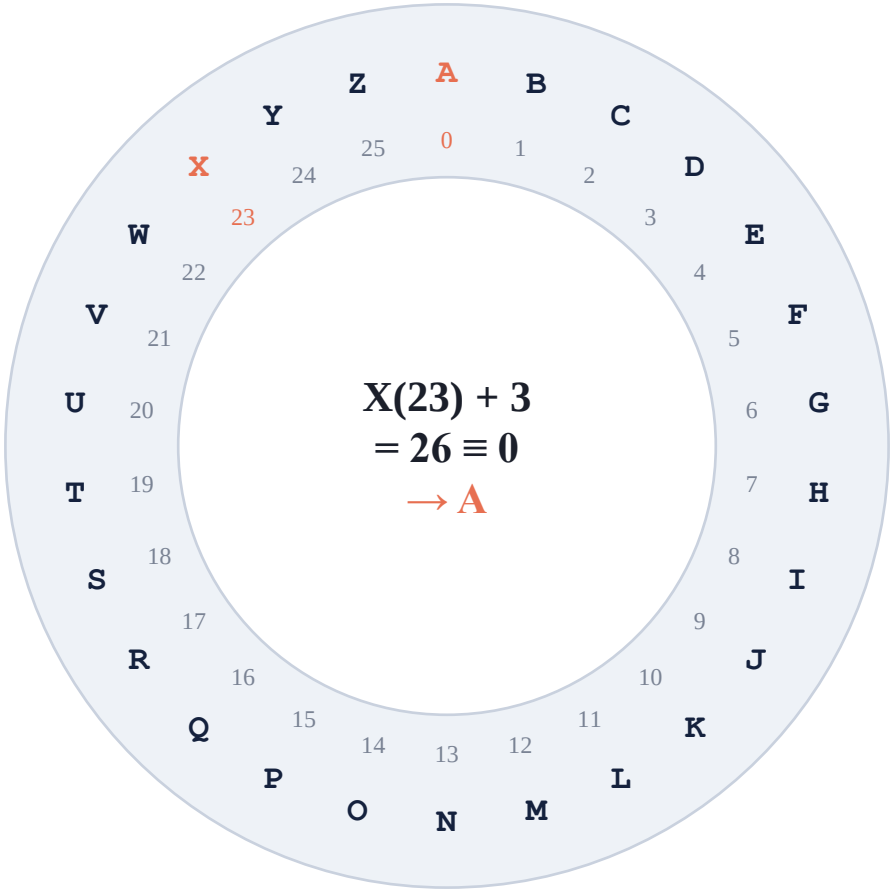
Kriptografiyada qayerda kerak

- Sezar, affin, Vijener — harf raqamlari ustida (7–10)
- XOR — Vernam, oqimli shifrlar, AES ichida (15–17, 25)
- Matn, rasm, ovoz — avval baytga aylanib, keyin shifrlanadi

Raqamlar

1 bayt = 8 bit; A = 65 = 0x41

Modul arifmetika: aylana ustidagi hisob



Qoldiq

a mod n — n ga bo‘lgandagi qoldiq: 0, 1, …, n − 1

Soat: 10 dan 5 soat o‘tsa → 15 mod 12 = 3

Taqqoslama

a ≡ b (mod n) — qoldiqlari bir xil

29 ≡ 3 (mod 26): 29-“harf” — bu D

Bo‘lish = teskariga ko‘paytirish

a^{-1} mavjud ⇔ EKUB(a, n) = 1

mod 26 da 12 ta: 1 3 5 7 9 11 15 17 19 21 23 25

Kriptografiyada qayerda kerak

- mod 26: Sezar, affin, Vijener (7–10)
- mod 2: XOR (2-mavzu)
- katta modul: RSA, Diffi–Xellman (29–30)
- affin kaliti a — faqat chapdagi 12 sondan

Evklid algoritmi: EKUB(a, b) = EKUB(b, a mod b)

Misol: EKUB(252, 105)

Qadam	Bo‘lish	Qoldiq
1	252 = 2 · 105 + 42	42
2	105 = 2 · 42 + 21	**21**
3	42 = 2 · 21 + 0	0

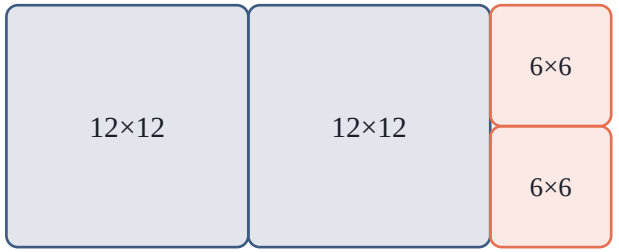
EKUB(252, 105) = 21

oxirgi noldan farqli qoldiq

Qoida

Kattasini kichigiga bo‘lamiz, so‘ng bo‘luvchi va qoldiq bilan takrorlaymiz. Qoldiq 0 bo‘lsa — to‘xtaymiz.

Geometrik ma’no



30 × 12: ikki kvadrat 12×12, qolgan 12×6 dan ikki 6×6 ⇒ EKUB = 6

Kriptografiyada qayerda kerak

- EKUB = 1 → affin kaliti yaroqli (7)
- teskari element topish (keyingi slayd)
- RSA kalitini tekshirish (29)

Asosiy kontrast

EKUB topish — juda oson (Evklid, ~2300 yillik algoritm).

Katta sonni tub ko‘paytuvchilarga yoyish — amalda imkonsiz (600 xonali son).

“Bir tomonga oson, teskari tomonga qiyin” — RSA va ochiq kalitli kriptografiyaning asosi (18, 28–29-mavzular).

7^{-1} mod 26 ni topamiz: Evklidni orqaga yuramiz

1. Oldinga: oddiy Evklid

Bo‘lish	Qoldiqni ifodalaymiz
$26 = 3 \cdot 7 + 5$	$5 = 26 - 3 \cdot 7$
$7 = 1 \cdot 5 + 2$	$2 = 7 - 5$
$5 = 2 \cdot 2 + 1$	**$1 = 5 - 2 \cdot 2$**

2. Orqaga: 1 ni faqat 26 va 7 orqali yozamiz

$$1 = 5 - 2 \cdot 2$$ *oxirgi qatordan*

$$1 = 5 - 2 \cdot (7 - 5) = 3 \cdot 5 - 2 \cdot 7$$ *2 o‘rniga qo‘ydik*

$$1 = 3 \cdot (26 - 3 \cdot 7) - 2 \cdot 7 = 3 \cdot 26 - 11 \cdot 7$$ *5 o‘rniga qo‘ydik*

$$7 \cdot (-11) \equiv 1 \pmod{26} \Rightarrow 7^{-1} \equiv -11 + 26 = 15$$

tekshirish: $7 \cdot 15 = 105 = 4 \cdot 26 + 1$

Bezu ayniyati (aslida teorema)

ax + by = EKUB(a, b) bo‘ladigan butun x, y mavjud. Evklidni orqaga yurish ularni topadi.

Kriptografiyada qayerda kerak

- affin deshifrlash: $x = a^{-1}(y - b)$ (7)
- Hill: kalit matritsa teskarisi (13)
- RSA: yopiq kalit $d = e^{-1} \bmod \varphi(n)$ (29)

Teskari yo‘q bo‘lsa

8^{-1} mod 26: 8 va 26 ikkalasi juft, EKUB = 2 — teskari yo‘q

```
>>> pow(7, -1, 26)   15
```

Teskari element amalda: affin shifr kalitini ochish

Affin shifr $y = ax + b \pmod{26}$. Raqib biladi: **E→C, T→Z**

1	$4a + b \equiv 2, \quad 19a + b \equiv 25$	E = 4 → C = 2; T = 19 → Z = 25
2	$15a \equiv 23$	ayiramiz — b yo‘qoladi
3	$a \equiv 15^{-1} \cdot 23 = 7 \cdot 23 = 161 \equiv 5$	$15^{-1} = 7$ (Evklid, oldingi slayd)
4	$b \equiv 2 - 4 \cdot 5 = -18 \equiv 8$	kalit (a, b) = (5, 8)

Xulosa

Ikki juft ma’lum harf affin shifrning butun kalitini ochadi (ma’lum ochiq matn hujumi, 1-mavzu).

EKUK va davr: M-209 shifr mashinasi

Ikki tishli g‘ildirak bir vaqtda boshlang‘ich holatga qachon qaytadi? → EKUK qadamda.

4 va 6 tish: EKUK = 12 (ko‘paytma 24 emas)

3 va 4 tish: EKUK = 12 = 3 · 4 — o‘zaro tub

M-209: 6 g‘ildirak — 26, 25, 23, 21, 19, 17 tish, o‘zaro tub ⇒

davr = 101 405 850

Davr uzun → kalit takrorlanmaydi → shifrni ochish qiyinlashadi (11, 17-mavzular).

To‘rt formula — to‘rt shifr

n^k

takrorli so‘zlar

Vijener kaliti, 5 harf: $26^5 \approx 1{,}2 \cdot 10^7$

n bitli kalit: 2^n — har bir bit ×2

$n!$

o‘rin almashtirish

monoalfavit kalit: $26! \approx 4 \cdot 10^{26} \approx 2^{88}$

to‘liq tanlab bo‘lmaydi (lekin 8-mavzu!)

$n!/(n-k)!$

tartib bilan tanlash, A(n, k)

Enigma: 5 rotordan 3 tasi tartib bilan

$A(5, 3) = 5 \cdot 4 \cdot 3 = 60$

$n!/(k!(n-k)!)$

tartibsiz tanlash, C(n, k)

Enigma paneli, bitta sim: $C(26, 2) = 325$

A–B va B–A — bitta ulanish

Dirixle prinsipi

n qutiga n + 1 narsa ⇒ kamida bitta qutida ikkita.

256 bitli xesh: qiymatlar 2^{256} ta, xabarlar esa cheksiz ⇒ kolliziyalar ALBATTA bor.

Xavfsizlik — ularning yo‘qligi emas, TOPISH qiyinligi (33-mavzu).

Enigma: $1{,}6 \cdot 10^{20}$ kalit qayerdan keladi

Rotorlar tartibi
$A(5, 3) = 60$

×

Boshlang‘ich holat
$26^3 = 17\,576$

×

Panel (10 juft sim)
$\approx 1{,}5 \cdot 10^{14}$

=

jami
$\approx 1{,}6 \cdot 10^{20}$

Enigma qanday ishlaydi

Tugma → panel → 3 rotor → reflektor → orqaga → lampochka.

Har tugmadan keyin rotor buriladi: AAAA → har safar boshqa harf.
Mazmunan — polialfavit almashtirish (10, 14-mavzular).

5000+ yil

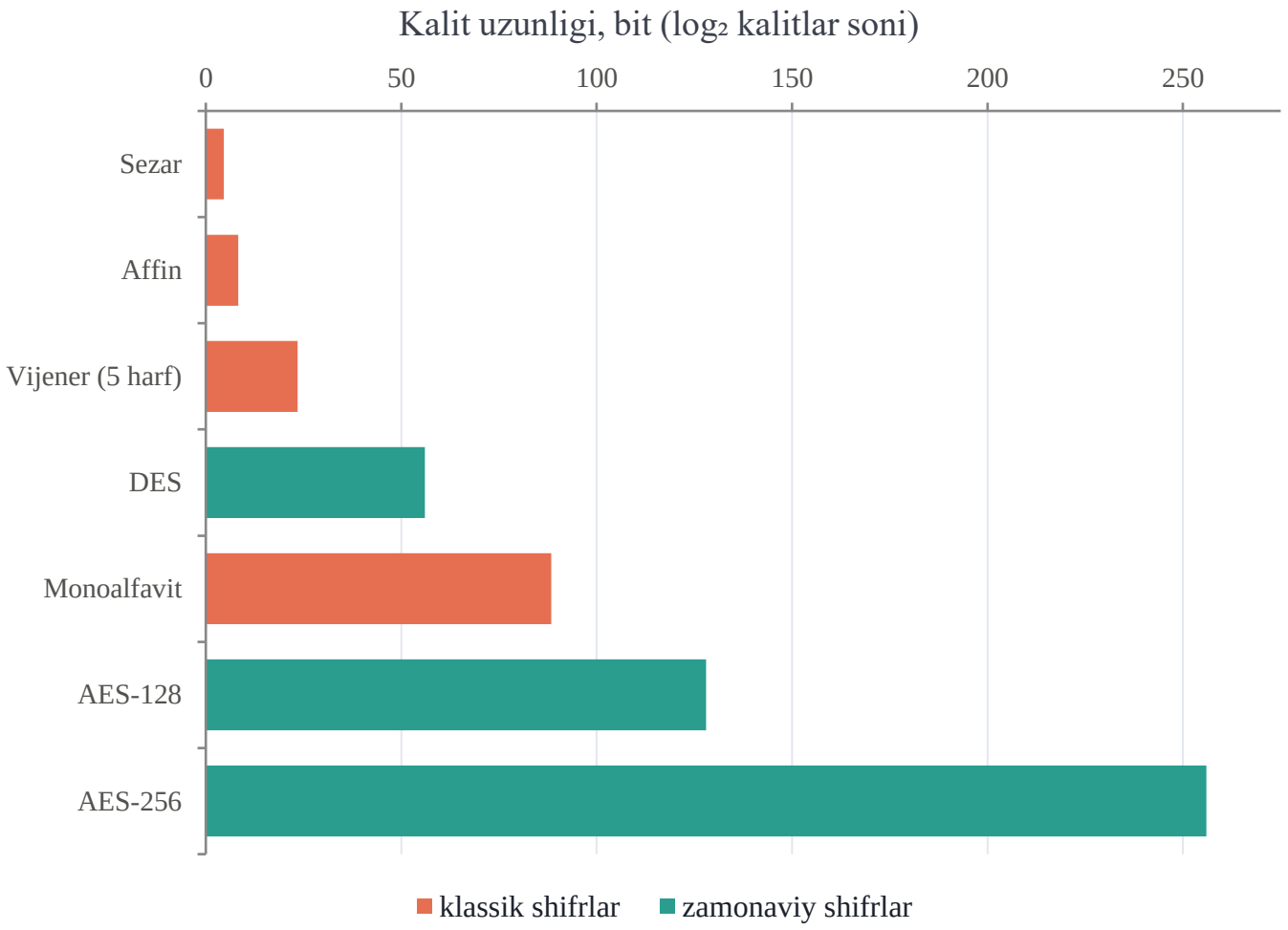
barcha kalitlarni 10^9 kalit/s tezlikda sinash vaqti

Lekin Enigma ochilgan — kalitlarni sinab emas

Zaiflik: harf hech qachon o‘ziga shifrlanmaydi + operator xatolari va standart so‘zlar (“kriblar”, masalan WETTERBERICHT — “ob-havo hisoboti”). Rejevskiy (1932) → Tyuring, “Bombe” (1940).

Kalitlar soni ≠ xavfsizlik.

Kalitlar fazosi: zarur, lekin yetarli emas



(N + 1) / 2

to‘liq tanlashda o‘rtacha urinishlar soni

56 soat

DES (2^{56}) ni EFF “Deep Crack” qurilmasi ochgan, 1998

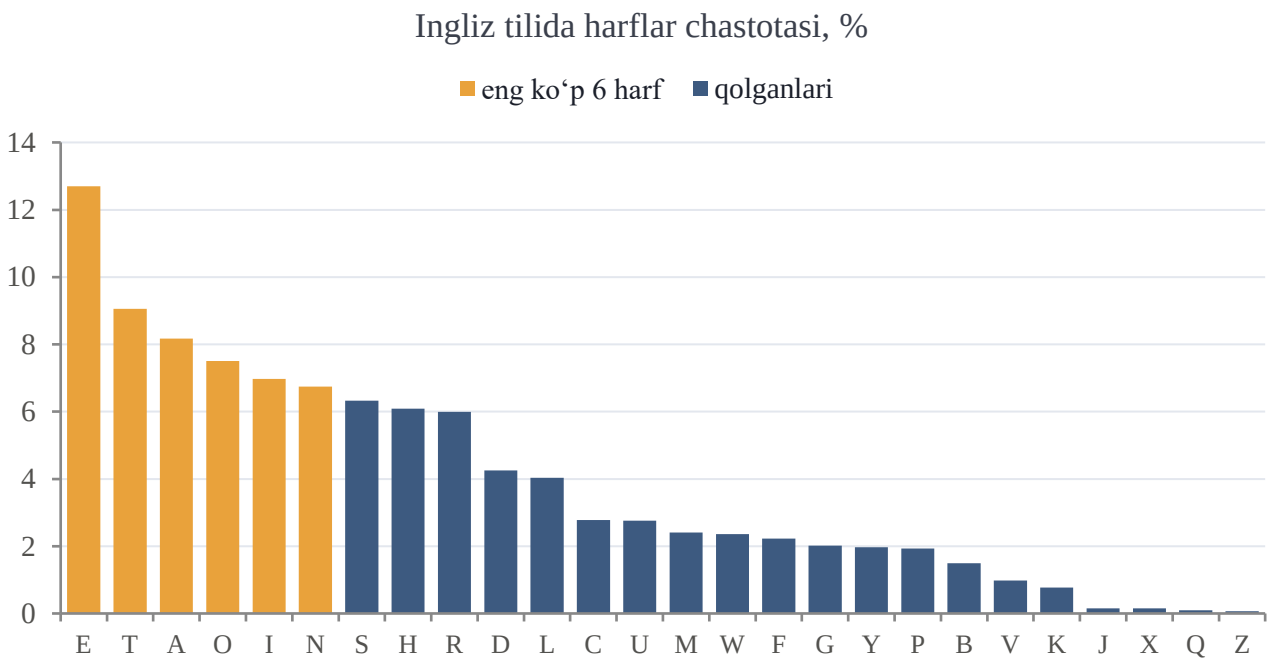
≈ 10^{19} yil

AES-128 — 10^{12} kalit/s tezlikda (koinot yoshi ≈ $1{,}4 \cdot 10^{10}$ yil)

Saboq

Monoalfavit ≈ 2^{88} > DES 2^{56}, lekin chastotali tahlil bilan daqiqalarda ochiladi → 6-mavzu.

Til tasodifiy emas: chastotali tahlil



Rus tilida: O 10,97 %, E 8,45 %, A 8,01 %; eng kamyob — Ъ va Ё (0,04 %)

χ^2 — ochilgan matn tilga qanchalik o‘xshash

Har kalit bilan ochib, harflar sonini ingliz tilidagi bilan solishtiramiz: farq eng kichik — to‘g‘ri kalit

Misol: Sezar shifrmatni (kalit noma’lum)

```
PHHW PH QHDU WKH HDVW JDWH DW VHYHQ
```

harf	H	W / D
sanoq (28 tadan)	**9**	5 / 4

$$H = E \Rightarrow K = 7 - 4 = 3$$

```
MEET ME NEAR THE EAST GATE AT SEVEN
```

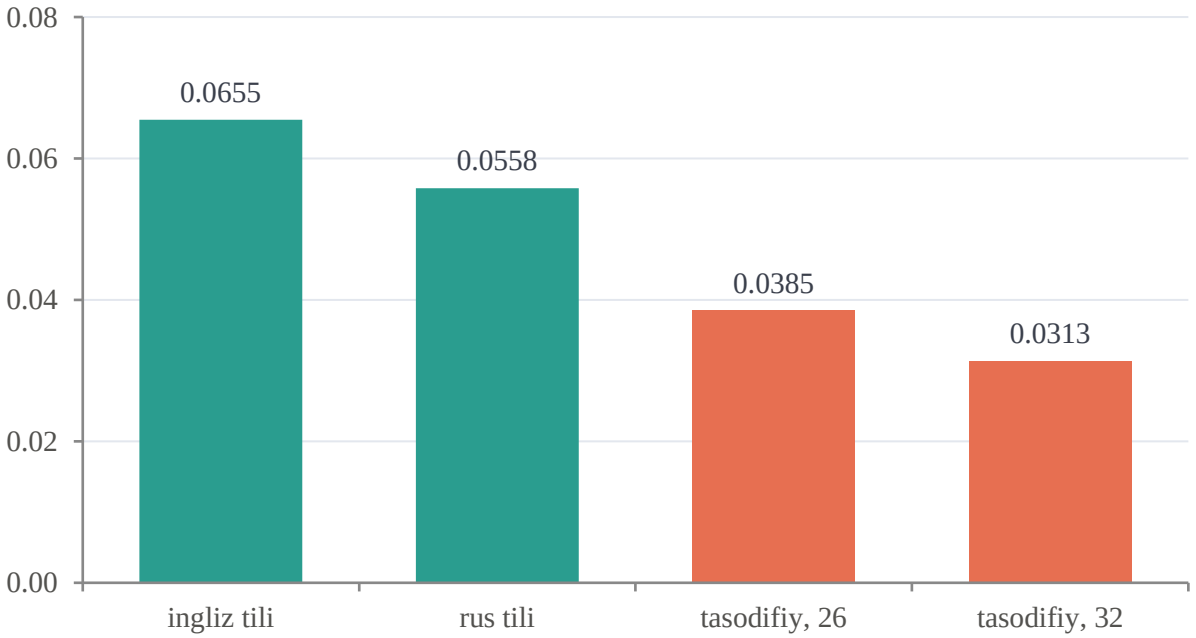
K	3	14	0
χ^2	**28,5**	75,0	274,2

Moslik indeksi: shifr turini aniqlash

Moslik indeksi (IC) — nima?

Matndan tasodifan olingan ikki harf bir xil chiqish ehtimolligi

inglizcha matn ≈ 0,066; tasodifiy harflar ≈ 0,038



1. Monoalfavit — IC o‘zgarmaydi

Almashtirish harflar nomini o‘zgartiradi, sanoqlarini emas: shifrmatnda ham IC ≈ 0,066.

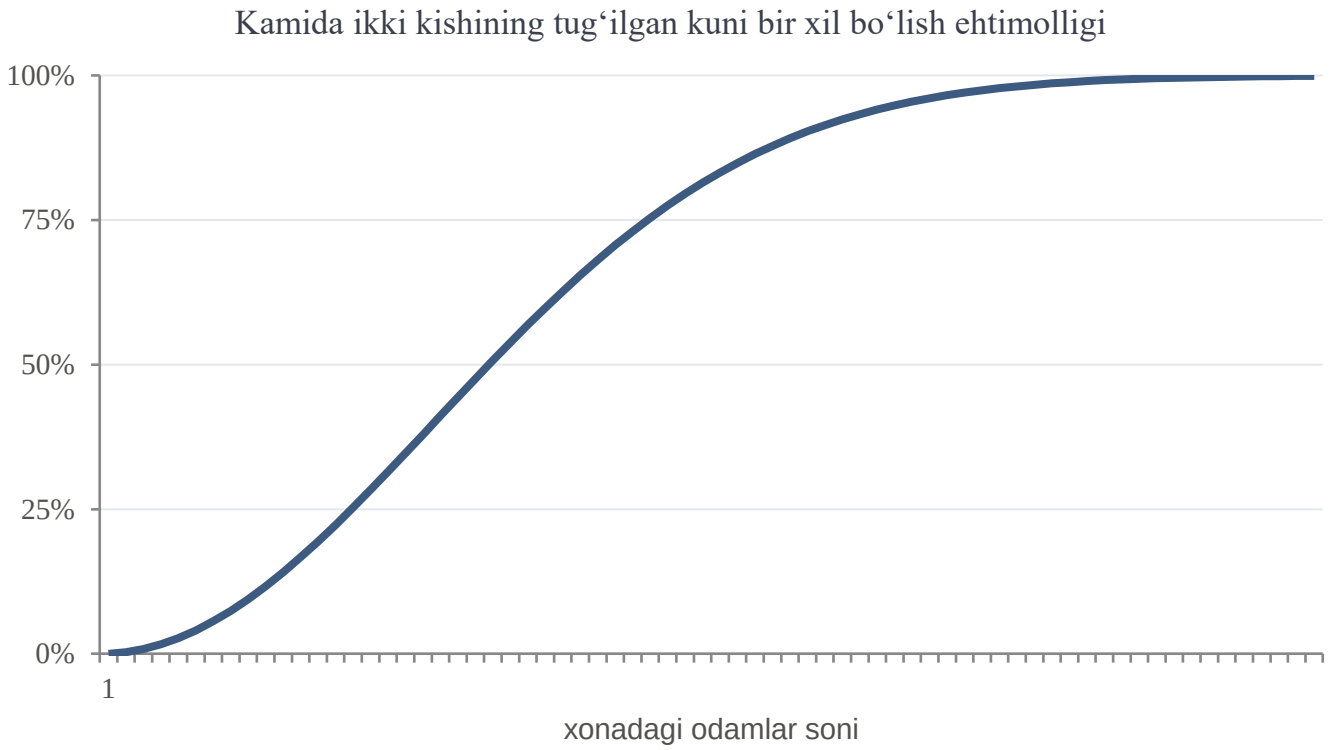
2. Polialfavit — IC pasayadi

Chastotalar tekislanadi: IC ≈ 0,038 ga yaqinlashadi → kalit uzunligini topish mumkin (Fridman testi, 11-mavzu).

Kriptografiyada qayerda kerak

- shifr turini aniqlash: monoalfavitmi yoki polialfavitmi
- Vijener kaliti uzunligini topish (11-mavzu)
- diqqat: kamida bir necha yuz harf kerak

Tug‘ilgan kun paradoksi: √N qoidasi



Hisoblash: hamma kunlar har xil bo‘lish ehtimolini topib, 1 dan ayiramiz

23 kishi

→ 50,7 % (ko‘pchilik 183 deydi)

Sabab: C(23, 2) = 253 juftlik — har biri mos kelish imkoniyati

N qiymat ⇒ ≈ √N tanlovdan keyin takrorlanish

Kriptografiyada: xesh kolliziyasi uchun urinishlar

xesh, bit	64	256
urinish ≈ $2^{n/2}$	2^{32}	2^{128}

Shuning uchun xesh kalitdan 2 baravar uzun tanlanadi (33-mavzu).

I blokdan II blokka: vosita → qo‘llanish

Modul arifmetika	→	Sezar, affin, Vijener — barchasi qo‘shish mod 26
Teskari element (Evklid)	→	affin rasshifrovkasi, Hill matritsasi, RSA yopiq kaliti
Kombinatorika	→	kalitlar fazosi va to‘liq tanlash bahosi
Chastota va χ^2	→	monoalfavit shifrni ochish (8-mavzu)
Moslik indeksi	→	Vijener kalit uzunligi — Fridman testi (11-mavzu)
√N qoidasi	→	xesh uzunligini tanlash (33-mavzu)

Keyingi mavzu: 7. Oddiy almashtirish shifrlari — Sezar, Atbash, affin